```
import pandas as pd
# basic data exploration
# 1. Load train.csv
df = pd.read_csv("/content/train.csv")
print(df)
```

```
     PassengerId  Survived  Pclass  \
0              1         0       3
1              2         1       1
2              3         1       3
3              4         1       1
4              5         0       3
..           ...       ...     ...
886          887         0       2
887          888         1       1
888          889         0       3
889          890         1       1
890          891         0       3

                                                 Name     Sex   Age  SibSp  \
0                             Braund, Mr. Owen Harris    male  22.0      1
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                              Heikkinen, Miss. Laina  female  26.0      0
3          Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                            Allen, Mr. William Henry    male  35.0      0
..                                                ...     ...   ...    ...
886                            Montvila, Rev. Juozas    male  27.0      0
887                     Graham, Miss. Margaret Edith  female  19.0      0
888         Johnston, Miss. Catherine Helen "Carrie"  female   NaN      1
889                            Behr, Mr. Karl Howell    male  26.0      0
890                              Dooley, Mr. Patrick    male  32.0      0

     Parch            Ticket     Fare Cabin Embarked
0        0         A/5 21171   7.2500   NaN        S
1        0          PC 17599  71.2833   C85        C
2        0  STON/O2. 3101282   7.9250   NaN        S
3        0            113803  53.1000  C123        S
4        0            373450   8.0500   NaN        S
..     ...               ...      ...   ...      ...
886      0            211536  13.0000   NaN        S
887      0            112053  30.0000   B42        S
888      2        W./C. 6607  23.4500   NaN        S
889      0            111369  30.0000  C148        C
890      0            370376   7.7500   NaN        Q

[891 rows x 12 columns]
```

```
# 2. First 10 rows
print(df.head(10))

# 3. Last 10 rows
print(df.tail(10))

# 4.  8 random rows
print(df.sample(8))
```

```
885  female  39.0      0      5            382652  29.1250   NaN        Q
886    male  27.0      0      0            211536  13.0000   NaN        S
887  female  19.0      0      0            112053  30.0000   B42        S
888  female   NaN      1      2        W./C. 6607  23.4500   NaN        S
889    male  26.0      0      0            111369  30.0000  C148        C
890    male  32.0      0      0            370376   7.7500   NaN        Q
     PassengerId  Survived  Pclass  \
678          679         0       3
779          780         1       1
881          882         0       3
364          365         0       3
534          535         0       3
765          766         1       1
811          812         0       3
801          802         1       2

                                                 Name     Sex   Age  SibSp  \
678            Goodwin, Mrs. Frederick (Augusta Tyler)  female  43.0      1
779  Robert, Mrs. Edward Scott (Elisabeth Walton Mc...  female  43.0      0
881                                Markun, Mr. Johann    male  33.0      0
364                              O'Brien, Mr. Thomas    male   NaN      1
534                              Cacic, Miss. Marija  female  30.0      0
765              Hogeboom, Mrs. John C (Anna Andrews)  female  51.0      1
811                                Lester, Mr. James    male  39.0      0
801        Collyer, Mrs. Harvey (Charlotte Annie Tate)  female  31.0      1

     Parch      Ticket      Fare Cabin Embarked
678      6     CA 2144   46.9000   NaN        S
779      1       24160  211.3375    B3        S
881      0      349257    7.8958   NaN        S
364      0      370365   15.5000   NaN        Q
534      0      315084    8.6625   NaN        S
765      0       13502   77.9583   D11        S
811      0   A/4 48871   24.1500   NaN        S
801      1  C.A. 31921   26.2500   NaN        S
```

```
# 5. Shape
print(df.shape)

# 6. Column names
print(df.columns)

# 7. Data types
print(df.dtypes)
```

```
(891, 12)
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
dtype: object
```

```
# 8. Info
df.info()

# 9. Statistics
print(df.describe())

# 10. Total rows and columns
rows, columns = df.shape
print("Total Rows:", rows)
print("Total Columns:", columns)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
```

```
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
Total Rows: 891
Total Columns: 12
```

```
# Missing Values Handling
# 11. Find total missing values in each column
print(df.isnull().sum())


# 12. Fill missing values in Age column with mean age
df["Age"] = df["Age"].fillna(df["Age"].mean())
print(df["Age"].isnull().sum())


# 13. Fill missing values in Embarked column with mode
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
print(df["Embarked"].isnull().sum())


# 14. Fill missing values in Cabin with 'Unknown'
df_cabin_unknown = df.copy()
df_cabin_unknown["Cabin"] = df_cabin_unknown["Cabin"].fillna("Unknown")
print(df_cabin_unknown["Cabin"].head())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
0
0
0    Unknown
1        C85
2    Unknown
3       C123
4    Unknown
Name: Cabin, dtype: object
```

```
# column section
# 18. Select Name, Sex, and Age columns
print(df[["Name", "Sex", "Age"]])


# 19. Select Pclass, Fare, and Survived columns
```

```
print(df[["Pclass", "Fare", "Survived"]])


# 20. Display only Ticket and Cabin columns
print(df[["Ticket", "Cabin"]])
```

```
                                                Name     Sex        Age
0                            Braund, Mr. Owen Harris    male  22.000000
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.000000
2                             Heikkinen, Miss. Laina  female  26.000000
3        Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.000000
4                           Allen, Mr. William Henry    male  35.000000
..                                               ...     ...        ...
886                            Montvila, Rev. Juozas    male  27.000000
887                     Graham, Miss. Margaret Edith  female  19.000000
888         Johnston, Miss. Catherine Helen "Carrie"  female  29.699118
889                            Behr, Mr. Karl Howell    male  26.000000
890                              Dooley, Mr. Patrick    male  32.000000

[891 rows x 3 columns]
     Pclass     Fare  Survived
0         3   7.2500         0
1         1  71.2833         1
2         3   7.9250         1
3         1  53.1000         1
4         3   8.0500         0
..      ...      ...       ...
886       2  13.0000         0
887       1  30.0000         1
888       3  23.4500         0
889       1  30.0000         1
890       3   7.7500         0

[891 rows x 3 columns]
               Ticket Cabin
0           A/5 21171   NaN
1            PC 17599   C85
2    STON/O2. 3101282   NaN
3              113803  C123
4              373450   NaN
..                ...   ...
886            211536   NaN
887            112053   B42
888        W./C. 6607   NaN
889            111369  C148
890            370376   NaN

[891 rows x 2 columns]
```

```
# Row Selection Using iloc[]

# 21. Display first 5 rows using iloc[]
print(df.iloc[:5])


# 22. Display rows 10 to 20
print(df.iloc[10:21])


# 23. Display rows 50 to 60 and columns Name, Age, Fare
print(df.iloc[50:61][["Name", "Age", "Fare"]])


# 24. Display last 5 rows using iloc[]
print(df.iloc[-5:])
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
```

```
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
    PassengerId  Survived  Pclass  \
10           11         1       3
11           12         1       1
12           13         0       3
13           14         0       3
14           15         0       3
15           16         1       2
16           17         0       3
17           18         1       2
18           19         0       3
19           20         1       3
20           21         0       2

                                            Name     Sex        Age  \
10                    Sandstrom, Miss. Marguerite Rut  female   4.000000
11                           Bonnell, Miss. Elizabeth  female  58.000000
12                     Saundercock, Mr. William Henry    male  20.000000
13                        Andersson, Mr. Anders Johan    male  39.000000
14                Vestrom, Miss. Hulda Amanda Adolfina  female  14.000000
15                    Hewlett, Mrs. (Mary D Kingcome)   female  55.000000
16                                Rice, Master. Eugene    male   2.000000
17                        Williams, Mr. Charles Eugene    male  29.699118
18  Vander Planke, Mrs. Julius (Emelia Maria Vande...  female  31.000000
19                              Masselmani, Mrs. Fatima  female  29.699118
20                                 Fynney, Mr. Joseph J    male  35.000000

    SibSp  Parch     Ticket     Fare Cabin Embarked
10      1      1    PP 9549  16.7000    G6        S
11      0      0     113783  26.5500  C103        S
12      0      0  A/5. 2151   8.0500   NaN        S
13      1      5     347082  31.2750   NaN        S
14      0      0     350406   7.8542   NaN        S
15      0      0     248706  16.0000   NaN        S
16      4      1     382652  29.1250   NaN        Q
17      0      0     244373  13.0000   NaN        S
18      1      0     345763  18.0000   NaN        S
19      0      0       2649   7.2250   NaN        C
20      0      0     239865  26.0000   NaN        S
```

```
# Row Selection Using loc[]

# 25. Display passengers whose age is greater than 60
print(df.loc[df["Age"] > 60])


# 26. Display passengers whose fare is greater than 100
print(df.loc[df["Fare"] > 100])


# 27. Display passengers whose sex is female
print(df.loc[df["Sex"] == "female"])


# 28. Display passengers who survived
print(df.loc[df["Survived"] == 1])


# 29. Display passengers in First Class
print(df.loc[df["Pclass"] == 1])
```

```
     PassengerId  Survived  Pclass                                     Name  \
33            34         0       2                    Wheadon, Mr. Edward H
54            55         0       1          Ostby, Mr. Engelhart Cornelius
96            97         0       1               Goldschmidt, Mr. George B
116          117         0       3                    Connors, Mr. Patrick
170          171         0       1               Van der hoef, Mr. Wyckoff
252          253         0       1               Stead, Mr. William Thomas
275          276         1       1       Andrews, Miss. Kornelia Theodosia
280          281         0       3                      Duane, Mr. Frank
326          327         0       3              Nysveen, Mr. Johan Hansen
438          439         0       1                     Fortune, Mr. Mark
456          457         0       1              Millet, Mr. Francis Davis
483          484         1       3                 Turkula, Mrs. (Hedwig)
493          494         0       1               Artagaveytia, Mr. Ramon
545          546         0       1            Nicholson, Mr. Arthur Ernest
555          556         0       1                    Wright, Mr. George
570          571         1       2                    Harris, Mr. George
625          626         0       1                  Sutton, Mr. Frederick
```

```
630          631         1       1       Barkworth, Mr. Algernon Henry Wilson
672          673         0       2               Mitchell, Mr. Henry Michael
745          746         0       1              Crosby, Capt. Edward Gifford
829          830         1       1  Stone, Mrs. George Nelson (Martha Evelyn)
851          852         0       3                      Svensson, Mr. Johan

        Sex   Age  SibSp  Parch       Ticket      Fare        Cabin Embarked
33     male  66.0      0      0   C.A. 24579   10.5000          NaN        S
54     male  65.0      0      1       113509   61.9792          B30        C
96     male  71.0      0      0    PC 17754   34.6542           A5        C
116    male  70.5      0      0       370369    7.7500          NaN        Q
170    male  61.0      0      0       111240   33.5000          B19        S
252    male  62.0      0      0       113514   26.5500          C87        S
275  female  63.0      1      0        13502   77.9583           D7        S
280    male  65.0      0      0       336439    7.7500          NaN        Q
326    male  61.0      0      0       345364    6.2375          NaN        S
438    male  64.0      1      4        19950  263.0000  C23 C25 C27        S
456    male  65.0      0      0        13509   26.5500          E38        S
483  female  63.0      0      0         4134    9.5875          NaN        S
493    male  71.0      0      0    PC 17609   49.5042          NaN        C
545    male  64.0      0      0          693   26.0000          NaN        S
555    male  62.0      0      0       113807   26.5500          NaN        S
570    male  62.0      0      0  S.W./PP 752   10.5000          NaN        S
625    male  61.0      0      0        36963   32.3208          D50        S
630    male  80.0      0      0        27042   30.0000          A23        S
672    male  70.0      0      0   C.A. 24580   10.5000          NaN        S
745    male  70.0      1      1    WE/P 5735   71.0000          B22        S
829  female  62.0      0      0       113572   80.0000          B28        S
851    male  74.0      0      0       347060    7.7750          NaN        S
     PassengerId  Survived  Pclass  \
27            28         0       1
31            32         1       1
88            89         1       1
118          119         0       1
195          196         1       1
215          216         1       1
258          259         1       1
268          269         1       1
269          270         1       1
297          298         0       1
```

```
# Unique and Frequency Analysis

# 30. Find all unique values in the Sex column
print(df["Sex"].unique())


# 31. Count number of male and female passengers
print(df["Sex"].value_counts())


# 32. Find all unique values in the Embarked column
print(df["Embarked"].unique())


# 33. Count number of passengers in each passenger class
print(df["Pclass"].value_counts())


# 34. Count how many passengers survived and did not survive
print(df["Survived"].value_counts())
```

```
['male' 'female']
Sex
male      577
female    314
Name: count, dtype: int64
['S' 'C' 'Q']
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
Survived
0    549
1    342
Name: count, dtype: int64
```

```
# Statistical Analysis

# 35. Find the mean age of passengers
print(df["Age"].mean())
```

```
# 36. Find the median fare
print(df["Fare"].median())


# 37. Find the mode of the Embarked column
print(df["Embarked"].mode()[0])


# 38. Find average fare paid by female passengers
print(df.loc[df["Sex"] == "female", "Fare"].mean())


# 39. Find median age of passengers who survived
print(df.loc[df["Survived"] == 1, "Age"].median())


# 40. Find mean fare of first-class passengers
print(df.loc[df["Pclass"] == 1, "Fare"].mean())
```

```
29.69911764705882
14.4542
S
44.47981783439491
29.69911764705882
84.1546875
```

```
# Using apply()

# 46. Convert all passenger names to uppercase
df["Name_upper"] = df["Name"].apply(lambda x: x.upper())
print(df["Name_upper"].head())


# 47. Find the length of each passenger name
df["Name_length"] = df["Name"].apply(lambda x: len(x))
print(df["Name_length"].head())


# 48. Round the Fare column to two decimal places
df["Fare_rounded"] = df["Fare"].apply(lambda x: round(x, 2))
print(df["Fare_rounded"].head())


# 49. Convert Sex column to title case (Male, Female)
df["Sex_title"] = df["Sex"].apply(lambda x: x.title())
print(df["Sex_title"].head())
```

```
0                              BRAUND, MR. OWEN HARRIS
1    CUMINGS, MRS. JOHN BRADLEY (FLORENCE BRIGGS TH...
2                               HEIKKINEN, MISS. LAINA
3         FUTRELLE, MRS. JACQUES HEATH (LILY MAY PEEL)
4                             ALLEN, MR. WILLIAM HENRY
Name: Name_upper, dtype: object
0    23
1    51
2    22
3    44
4    24
Name: Name_length, dtype: int64
0     7.25
1    71.28
2     7.92
3    53.10
4     8.05
Name: Fare_rounded, dtype: float64
0      Male
1    Female
2    Female
3    Female
4      Male
Name: Sex_title, dtype: object
```

```
# Using map()

# 50. male -> M, female -> F
df["Sex_mapped"] = df["Sex"].map({"male": "M", "female": "F"})
print(df["Sex_mapped"].head())
```

```
# 51. 0 -> Did Not Survive, 1 -> Survived
df["Survival_status"] = df["Survived"].map({0: "Did Not Survive", 1: "Survived"})
print(df["Survival_status"].head())


# 52. Pclass mapping
df["Class_mapped"] = df["Pclass"].map({
    1: "First Class",
    2: "Second Class",
    3: "Third Class"
})
print(df["Class_mapped"].head())
```

```
0    M
1    F
2    F
3    F
4    M
Name: Sex_mapped, dtype: object
0    Did Not Survive
1           Survived
2           Survived
3           Survived
4    Did Not Survive
Name: Survival_status, dtype: object
0    Third Class
1    First Class
2    Third Class
3    First Class
4    Third Class
Name: Class_mapped, dtype: object
```